Experiment 21

MySQL String Functions – SPACE, SUBSTR, UPPER, LOWER, TRIM, LENGTH

Aim

To execute MySQL string functions that insert spaces, extract substrings, change letter case, remove surrounding spaces, and calculate string length.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program

Step 1: Create database

```
CREATE DATABASE LibraryDB;
USE LibraryDB;
```

Step 2: Create table and insert data

```
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    book_name VARCHAR(50)
);

INSERT INTO Books VALUES
(1, 'Database'),
(2, 'Computer'),
(3, 'Programming');
```

Step 3: Display table

```
SELECT * FROM Books;
```

Queries

SPACE()

```
SELECT CONCAT('SQL', SPACE(3), 'QUERY');
```

SUBSTR()

```
SELECT SUBSTR('LIBRARY', 1, 4);
```

UPPER()

```
SELECT UPPER('library');
```

LOWER()

```
SELECT LOWER('MYSQL');
```

TRIM()

```
SELECT TRIM('   BOOKS   ');
```

LENGTH()

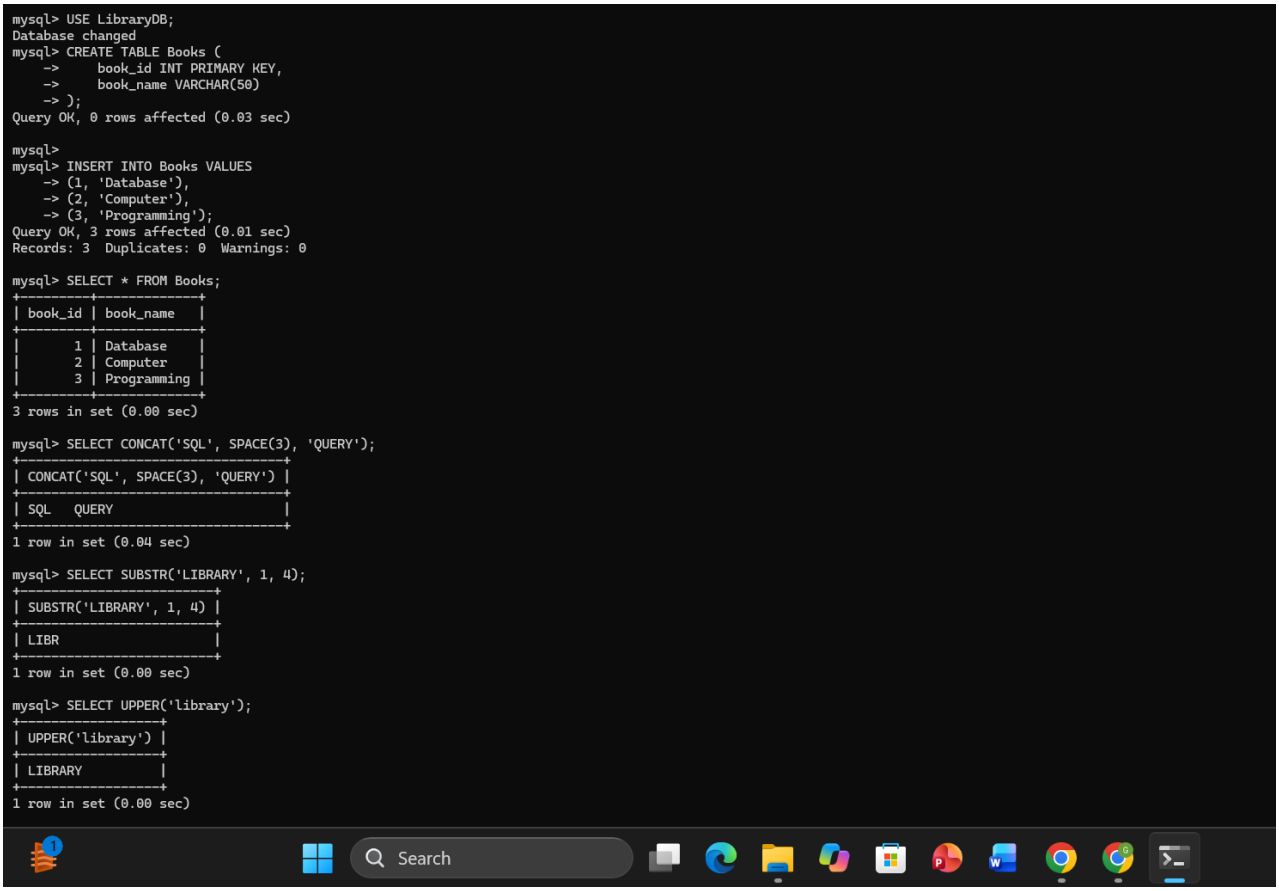
```
SELECT LENGTH('Library');
```

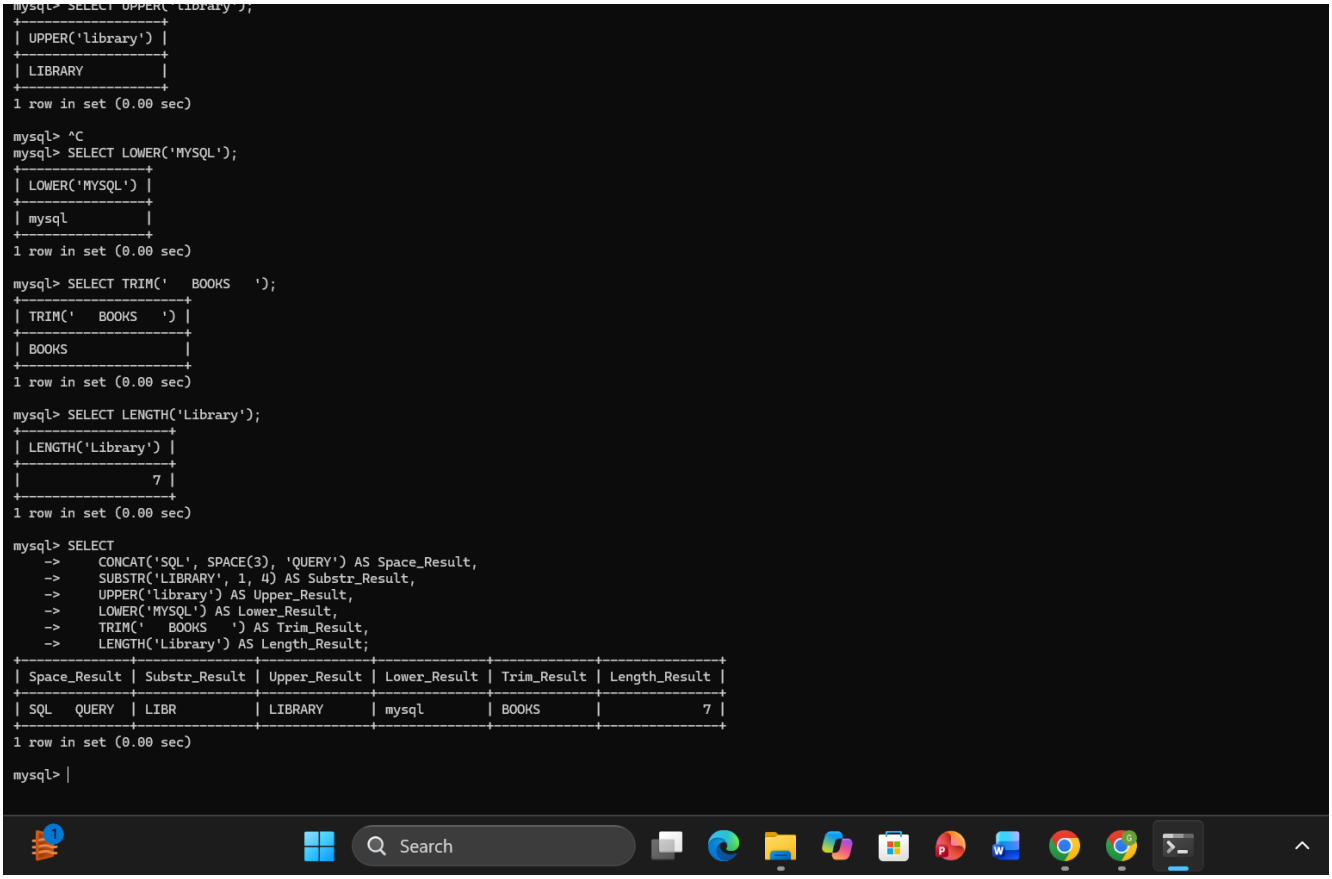
All functions in one query

```
SELECT
  CONCAT('SQL', SPACE(3), 'QUERY') AS Space_Result,
```

```
SUBSTR('LIBRARY', 1, 4) AS Substr_Result,
UPPER('library') AS Upper_Result,
LOWER('MYSQL') AS Lower_Result,
TRIM('    BOOKS    ') AS Trim_Result,
LENGTH('Library') AS Length_Result;
```

Expected Output:





Observation

The functions add spaces, extract part of a string, change case, trim surrounding spaces, and calculate string length in bytes.

Result

Thus, SPACE(), SUBSTR(), UPPER(), LOWER(), TRIM(), and LENGTH() were executed successfully.